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Imbalanced View Contribution Evaluation and Refinement for Deep Incomplete Multi-View Clustering

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Abstract

*In real-world applications, multi-view data often suffer from missing situations due to privacy protection and sensor failures. Such incomplete scenarios not only reduce information availability but also cause significant imbalance among views: certain "strong views" dominate the fusion process, while "weak views" contribute marginally, undermining cross-view collaboration. Existing incomplete multi-view clustering methods mainly focus on handling missing data, but overlook the view contribution imbalance induced by incompleteness and its impact on representation learning and clustering performance. To address this issue, we first analyze the data imbalance caused by missing views and the resulting disparities in view learning quality. Then, we propose an Imbalanced Contribution Evaluation and Refinement framework (**ICER**) for imbalanced incomplete multi-view clustering. Specifically, we employ Shapley values to quantify each view's marginal contribution, and incorporate unbalanced optimal transport to characterize distributional deviations across views. On this basis, we construct the view contribution imbalance metric to comprehensively evaluate cross-view collaboration and fusion quality, and design a collaboration enhancement module to explicitly reinforce inter-view cooperative optimization. Extensive experiments on multiple datasets demonstrate that the proposed method outperforms existing incomplete multi-view clustering approaches, validating the effectiveness and necessity of explicitly modeling and mitigating view imbalance in imbalanced incomplete scenarios. Code is released at https://github.com/Evelyn-zhou24/ICER.*

1. Introduction

Multi-view data, which integrates complementary information from multiple sensors or heterogeneous sources, has demonstrated significant value in tasks such as classification, clustering, and retrieval [6, 17, 25, 36, 51]. Thanks to recent advances in data acquisition and feature extraction technologies, obtaining multi-view data has become increasingly convenient. However, in real-world applications, some views are often missing due to privacy protection, sensor failure, or transmission errors, leading to incomplete multi-view data [30, 48]. Existing multi-view clustering (MVC) methods [7, 10, 24] generally assume that all views are completely observable, making them unsuitable for such incomplete scenarios. To address this issue, researchers have proposed a wide range of Incomplete Multi-view Clustering (IMVC) methods [18, 23, 40]. Traditional IMVC approaches are mainly based on matrix factorization[11, 45],

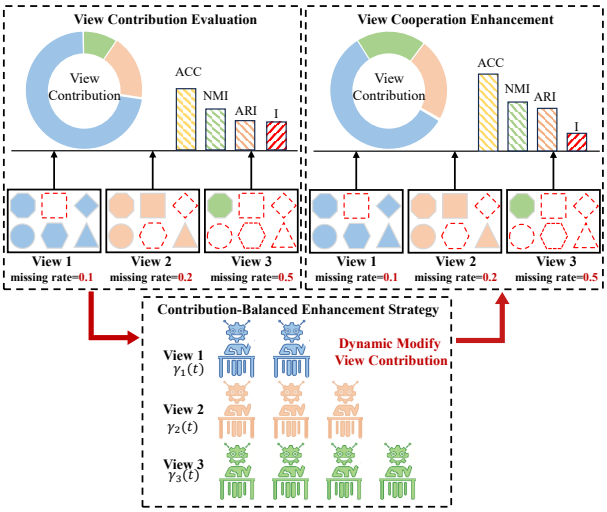


Figure 1. Illustration of the imbalance situation in incomplete multi-view clustering. The upper part shows view contribution evaluation and cooperation enhancement across views with different missing rates. The lower part depicts a view-adaptive curriculum learning strategy that dynamically adjusts view contributions to mitigate imbalance. γ denotes the curriculum learning progress.

kernel learning[26, 38], graph learning [2, 37], or subspace learning techniques [31, 49]. However, due to their inherent linear assumptions, these methods struggle to effectively capture the complex nonlinear structures in real data. In contrast, deep IMVC (DIMVC) methods leverage nonlinear representation learning to achieve substantial performance improvements and have thus become a research hotspot in recent years [8, 44].

Despite the remarkable progress achieved by deep methods [4, 14], most existing studies still implicitly assume that all views contribute equally during feature fusion. This assumption is often invalid in practice — the randomness and imbalance of missing data lead to significant disparities in both the observability and feature quality across views. Ignoring such missing-induced imbalance tends to overestimate the influence of "strong views" while underestimating the value of "weak views," causing local features to be suppressed, cross-view relationships to be distorted, and ultimately resulting in unstable training and degraded clustering performance. Essentially, missingness is not merely an issue of data incompleteness, but a fundamental mechanism that causes imbalance in cross-view collaboration.

This leads to two key scientific questions: (1) How can we quantify the marginal contribution of each view under imbalanced missing conditions? The heterogeneous observation ratios and feature qualities across views make it challenging to establish a unified and comparable metric for evaluating their collaborative contributions; (2) How can we adaptively enhance cross-view collaboration based on these evaluations? The model needs to regulate the influence of strong and weak views to alleviate dominance and insufficient participation during the fusion process.

To address these challenges, as depicted in Fig. 1, we propose an **I**mbalanced view **C**ontribution **E**valuation and **R**efinement (ICER) framework for deep incomplete multi-view clustering, which explicitly evaluates and mitigates view contribution imbalance induced by imbalanced missing conditions. First, we employ the Shapley value from cooperative game theory to quantify the marginal contribution of each view to the clustering objective, providing cross-view comparable contribution measures. Second, to model the mismatch across views caused by both missingness and distributional shifts, we introduce Unbalanced Optimal Transport (UOT) and define a new metric called view contribution imbalance to comprehensively evaluate the degree of cross-view collaboration and fusion quality. Based on this evaluation, we design a collaboration enhancement module that incorporates the assessment results as priors within a view-adaptive curriculum learning mechanism. This allows the model to explicitly suppress the dominance of strong views, enhance the participation of weak views, and promote overall collaborative optimization.

The main contributions of this paper can be summarized as follows:

- We provide the first systematic analysis of view imbalance arising from missing data, and show how it substantially affects cross-view collaboration and clustering outcomes.
- By leveraging the Shapley value and unbalanced optimal transport, we define the view contribution imbalance metric, and further design a collaboration enhancement module that dynamically strengthens weak views through curriculum learning, enabling adaptive optimization at the collaboration level.
- Extensive experiments on five public multi-view benchmark datasets demonstrate that our method consistently outperforms existing approaches under imbalanced missing conditions, validating the effectiveness and necessity of explicitly modeling and alleviating view imbalance in multi-view clustering.

2. Related Work

2.1. Incomplete Multi-view Clustering

Existing incomplete multi-view clustering (IMVC) methods can be broadly categorized into traditional and deep approaches. Traditional methods typically pre-process missing views using zero/mean filling or simple interpolation [34], followed by classical multi-view learning techniques such as non-negative matrix factorization [16, 35], kernel learning [21, 22], and graph-based models [5, 32], among others. However, these approaches often rely on shallow architectures or matrix inversion, limiting their scalability [19]. More importantly, they usually treat missing data as homogeneous noise while overlooking that missing patterns may vary significantly across views, leading to biased information completeness and unequal view contributions during feature fusion.

With the advancement of deep neural networks, DIMVC methods have achieved significant progress by leveraging nonlinear representation learning to infer latent embeddings for missing data. Representative studies include distribution alignment without explicit imputation [41], GAN-based data generation with adaptive fusion [29, 33], and cross-view contrastive recovery of missing semantics [15]. Despite their success, these models generally assume equal contributions among views once data are reconstructed, and fail to account for the view imbalance induced by non-uniform missingness. Such imbalance, if not properly modeled, can distort fusion representations and degrade clustering performance. Addressing this overlooked issue is the key motivation of our work.

2.2. Imbalanced Multi-view Learning

Imbalanced multi-view learning has recently attracted increasing attention in multi-modal representation analysis.

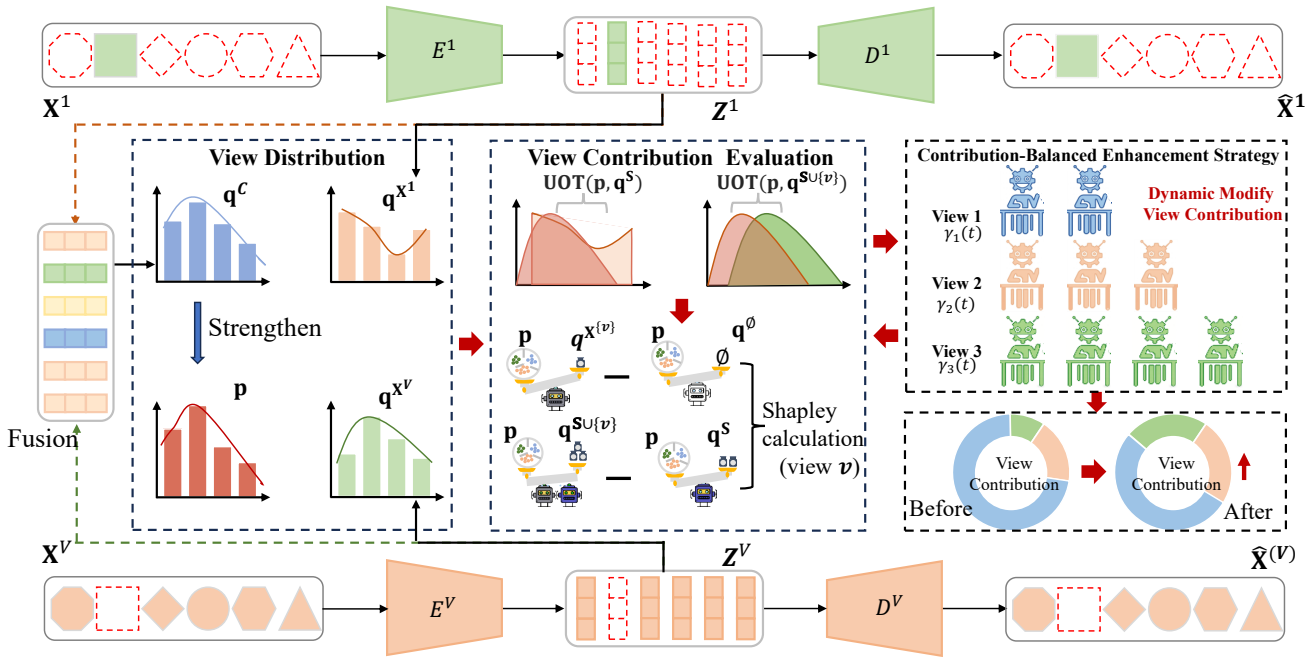


Figure 2. The overall framework of ICER. ICER mainly consists of two main components: (1) the View Contribution Evaluation module, which estimates each view's marginal contribution using Shapley values based on UOT-derived discrepancies between the strengthened fusion distribution and view-specific distributions; and (2) the Contribution-Balanced Enhancement module, which dynamically adjusts the learning pace of each view to alleviate contribution imbalance. By iteratively applying these two modules during training, ICER achieves balanced cross-view collaboration under imbalanced incomplete multi-view settings.

Some studies address view-quality discrepancies through adaptive weighting or attention mechanisms that suppress noisy views while emphasizing more informative ones [42, 50]. Other works alleviate cross-view information imbalance via contrastive or collaborative optimization strategies [12, 20]. Wang *et al.* [28] further proposed a cooperation enhancement framework that evaluates the cooperative value of each view from a game-theoretic perspective and strengthens inter-view collaboration. Inspired by these works, we further investigate the problem of imbalanced incomplete multi-view clustering. Such imbalance is reflected not only in the heterogeneous data observability across views, but also in the unequal contributions of different views to the clustering objective during feature fusion. Existing methods rarely provide a unified framework to jointly model missing data and view contribution imbalance. To address this gap, we propose a framework for evaluating and enhancing view cooperation under imbalanced missing conditions. Specifically, the framework leverages Shapley values and an imbalance-aware collaborative learning mechanism to regulate cooperative learning dynamics.

3. Method

In this section, we first present the overall architecture of ICER, as illustrated in Fig. 2. We then introduce each component of the proposed framework in detail, followed by the objective function and the corresponding optimization procedure.

3.1. Problem Statement

Given an incomplete multi-view dataset $\{\mathbf{X}^v \in \mathbb{R}^{N\times d_v}\}_{v=1}^{V}$, where V denotes the number of views, d_v is the feature dimension of the v-th view, and N is the total number of samples. To indicate the presence of missing data, we define an indicator matrix $\mathbf{A} = [a_i^v] \in \{0,1\}^{N\times V}$, where $a_i^v = 0$ if the i-th sample in the v-th view is missing, and $a_i^v = 1$ otherwise.

3.2. Incomplete Multi-view Feature Learning

We equip each view with an autoencoder to learn clustering-friendly feature representations. For the v-th view, an encoder–decoder pair, denoted as E^v and D^v, is used to obtain the high-level embedding representation $\mathbf{Z}^v$. Specifically, the feature extraction process is formulated as: $E^v(\mathbf{X}^v;\theta^v) : \mathbf{X}^v \in \mathbb{R}^{N\times d_v} \rightarrow \mathbf{Z}^v \in \mathbb{R}^{N\times d}$, and the feature reconstruction process by the decoder D^v is given as: $D^v(\mathbf{Z}^v;\phi^v) : \mathbf{Z}^v \in \mathbb{R}^{N\times d} \rightarrow \hat{\mathbf{X}}^v \in \mathbb{R}^{N\times d_v}$, where d denotes the dimension of the embedding space, and θ^v and ϕ^v are the learnable parameters of the v-th view's autoencoder.

The autoencoder projects features of each view into a common latent space, enabling the modeling of cross-view

distributions in a shared space. The reconstruction loss for all views is defined as:

$$\begin{aligned}\mathcal{L}_{rec} &= \sum_{v=1}^{V} \|\mathbf{X}^v - D^v(\mathbf{Z}^v)\|_F^2 \\ &= \sum_{v=1}^{V}\sum_{i=1}^{N} a_i^v \left\|\mathbf{x}_i^v - D^v(E^v(\mathbf{x}_i^v))\right\|_2^2 .\end{aligned} \tag{1}$$

In the pre-training stage, the model is optimized based on the above objective, where each view is still trained independently.

3.3. View Distribution Estimation

In incomplete multi-view scenarios, to obtain the clustering prediction results, each view employs a parameterized mapping to produce a soft cluster assignment matrix $\mathbf{Q}^v$, defined as

$$\mathcal{M}_v(\mathbf{Z}^v; \mathbf{U}^v) : \mathbf{Z}^v \in \mathbb{R}^{N\times d} \mapsto \mathbf{Q}^v \in \mathbb{R}^{N\times K}, \tag{2}$$

where $\mathbf{U}^v = [\mathbf{u}_1^v, \mathbf{u}_2^v, \ldots, \mathbf{u}_K^v] \in \mathbb{R}^{K\times d}$ represents the learnable cluster prototypes for the v-th view.

Specifically, for samples that are present in the v-th view, their cluster assignment distribution is defined as

$$q_{ij}^v = \begin{cases} \dfrac{(1+\|\mathbf{z}_i^v - \mathbf{u}_j^v\|_2^2)^{-1}}{\sum_{j'=1}^{K}(1+\|\mathbf{z}_i^v - \mathbf{u}_{j'}^v\|_2^2)^{-1}}, & a_i^v = 1, \\ 0, & a_i^v = 0, \end{cases} \tag{3}$$

where a_i^v is a binary visibility indicator, with $a_i^v = 1$ indicating that the i-th sample is available in the v-th view, and $a_i^v = 0$ indicating that it is missing. For the missing samples in a given view, a masking operation is applied accordingly.

In the v-th view, $\mathbf{u}_j^v$ denotes the j-th cluster centroid and q_{ij}^v is the soft assignment of embedding $\mathbf{z}_i^v$ to cluster j. This follows the end-to-end soft clustering mechanism of DEC [9, 39], which we extend to the incomplete multi-view setting.

We model incomplete multi-view clustering as an unsupervised cooperative game, where each view is treated as a player. Under the Shapley value framework, any subset of views $\mathcal{S} \subseteq \{1, \ldots, V\}$ forms a coalition, and each coalition corresponds to the fused distribution $\mathbf{Q}^{(\mathcal{S})}$ derived from the views it contains.

For the i-th sample, its fused clustering distribution under coalition $\mathcal{S}$ is defined as:

$$q_{ij}^{(\mathcal{S})} = \frac{\sum_{v\in\mathcal{S}} a_i^v \, \kappa_j^v \left(1+\|\mathbf{z}_i^v - \mathbf{u}_j^v\|_2^2\right)^{-1}}{\sum_{j'=1}^{K}\sum_{v'\in\mathcal{S}} a_i^{v'} \, \kappa_{j'}^{v'} \left(1+\|\mathbf{z}_i^{v'} - \mathbf{u}_{j'}^{v'}\|_2^2\right)^{-1}}. \tag{4}$$

Here, κ_j^v denotes the learnable weight of the v-th view for cluster j, which is normalized through the parameter matrix $\mathbf{W} \in \mathbb{R}^{K\times V}$ as:

$$\kappa_j^v = \frac{\exp(W_j^{(v)})}{\sum_{u\in\mathcal{S}} \exp(W_j^{(u)})}. \tag{5}$$

Moreover, when the coalition $\mathcal{S} = \emptyset$, it indicates that no views participate in the clustering process. In this case, the model cannot obtain any discriminative information from any view. Therefore, the clustering distribution of each sample is regarded as an uninformative uniform prior. In other words, for each sample i, its membership probabilities across all clusters are equal, i.e., $\mathbf{q}_i^{(\emptyset)} \sim \text{Uniform}(K)$, denotes the uniform distribution over the cluster set K, $\mathbf{Q}^{(\emptyset)} = \frac{1}{K}\mathbf{1}_{N\times K}$.

We regard the self-distilled and sharpened fused distribution $\mathbf{p}$ as the ideal trend of the global clustering structure. For any view coalition $\mathcal{S}$, its corresponding fused distribution $\mathbf{q}^{(\mathcal{S})}$ is obtained by aggregating information only from the actually visible views. The closer the coalition distribution $\mathbf{q}^{(\mathcal{S})}$ is to the target distribution $\mathbf{p}$, the greater contribution this coalition makes to reconstructing the complete clustering pattern.

$$p_{ij} = \frac{(q_{ij})^2}{\sum_j (q_{ij})^2}. \tag{6}$$

To quantify the consistency between the coalition-specific fused distribution $\mathbf{q}^{(\mathcal{S})}$ and the ideal global distribution $\mathbf{p}$ under missing views, we adopt the unbalanced optimal transport distance as a discrepancy measure.

Formally, given two discrete distributions $\mathbf{p}$ and $\mathbf{q}^{(\mathcal{S})}$ over K clusters, the UOT distance based on KL-relaxed marginals is defined as

$$\begin{aligned}\text{UOT}\big(\mathbf{p}, \mathbf{q}^{(\mathcal{S})}\big) = \min_{\pi\geq 0} \Big\{ &\langle G_{kl}, \pi_{kl}\rangle + \text{KL}\big(\pi\mathbf{1} \,\|\, \mathbf{p}\big) \\ &+ \text{KL}\big(\pi^\top\mathbf{1} \,\|\, \mathbf{q}^{(\mathcal{S})}\big)\Big\},\end{aligned} \tag{7}$$

where $G_{k\ell} = \|\mathbf{u}_k - \mathbf{u}_\ell\|_2^2$ denotes the transport cost between the k-th and ℓ-th cluster centroids in the shared latent space, $\pi \in \mathbb{R}^{K\times K}$ is the transport plan, and $\pi_{k\ell}$ represents the amount of mass transported from the k-th source cluster to the ℓ-th target cluster. By introducing the KL penalty terms on the row and column marginals, UOT [27] allows for non-conserved mass, which naturally accommodates the uncertainty and sample-size mismatch induced by missing views.

Based on this discrepancy, we define the utility of a coalition $\mathcal{S}$ as

$$E(\mathcal{S}) = \exp\big(-\text{UOT}(\mathbf{p}, \mathbf{q}^{(\mathcal{S})})\big), \tag{8}$$

When the optimal transport distance is small, the coalition-specific fused distribution $\mathbf{q}^{(\mathcal{S})}$ is closer to the ideal global trend $\mathbf{p}$, leading to a larger utility value $E(\mathcal{S})$.

To stabilize the global fusion while still encouraging each visible view to align with the consensus semantics, we introduce a cross-view consistency loss, which enforces the per-view clustering distributions to stay close to the final fused distribution in the semantic space:

$$\mathcal{L}_{\mathrm{ccl}} = \frac{1}{N}\sum_{v=1}^{V}\sum_{i=1}^{N} a_i^v \,\mathrm{UOT}\Big(\mathbf{q}_i^{(v)},\, \mathbf{q}_i^{(S)}\Big), \tag{9}$$

where $\mathbf{q}_i^{(v)}$ denotes the clustering distribution of the i-th sample in the v-th view, $\mathbf{q}_i^{(s)}$ denotes the final consensus distribution for that sample, and a_i^v is the observation indicator ensuring that only available views contribute to the loss.

3.4. View Contribution Evaluation

Building upon previous sections, we now quantify the marginal contribution of each view by means of the Shapley value, which provides a fair allocation of the coalition utility among all participating views.

Formally, the Shapley value of the v-th view is defined as

$$\phi^v = \sum_{\mathcal{S}\subseteq\mathcal{V}\setminus\{v\}} \frac{|\mathcal{S}|!\,(V-|\mathcal{S}|-1)!}{V!} \times \Big[E(\mathcal{S}\cup\{v\}) - E(\mathcal{S})\Big], \tag{10}$$

where $\mathcal{V}=\{1,2,\ldots,V\}$ denotes the set of all views, and $\mathcal{S}$ represents any subset that excludes view v. Substituting $E(\mathcal{S}) = \exp\big(-\mathrm{UOT}(\mathbf{p},\mathbf{q}^{(\mathcal{S})})\big)$, the above expression can be equivalently written as

$$\phi^v = \sum_{\mathcal{S}\subseteq\mathcal{V}\setminus\{v\}} \frac{|\mathcal{S}|!\,(V-|\mathcal{S}|-1)!}{V!} \times \Big[\exp\big(-\mathrm{UOT}(\mathbf{p},\mathbf{q}^{(\mathcal{S}\cup\{v\})})\big) - \exp\big(-\mathrm{UOT}(\mathbf{p},\mathbf{q}^{(\mathcal{S})})\big)\Big]. \tag{11}$$

The total contribution of all views equals the difference in utility between the full coalition and the empty coalition:

$$\sum_{v=1}^{V}\phi^v = E(\mathcal{V}) - E(\emptyset) = \exp\big(-\mathrm{UOT}(\mathbf{p},\mathbf{q}^{(\mathcal{V})})\big) - \exp\big(-\mathrm{UOT}(\mathbf{p},\mathbf{q}^{(\emptyset)})\big), \tag{12}$$

where $\mathbf{q}^{(\mathcal{V})}$ denotes the fused distribution obtained using all views, and $\mathbf{q}^{(\emptyset)}$ represents the uniform prior distribution when no view is available. During training, as $\mathbf{p}$ and $\mathbf{q}^{(\mathcal{V})}$ evolve, the Shapley values $\{\phi^v\}$ of all views will dynamically fluctuate. To measure the relative contribution of each view with respect to the overall fusion, we define the normalized Shapley value as $\psi^v = \frac{\phi^v}{\sum_{i=1}^{V}\phi^i}$, which reflects the proportional importance of each view in collaborative clustering.

Definition 1 (Global contribution-imbalance metric) *In imbalanced missing scenarios, the visible sample ratio, information completeness, and collaborative strength vary significantly across views, leading to structurally biased marginal contributions ψ^v. To quantitatively measure the overall contribution imbalance, we define a global imbalance index based on the normalized Shapley vector $\boldsymbol{\psi} = (\psi^1, \psi^2, \ldots, \psi^V)$, drawing on the concept of the Gini coefficient:*

$$I_\psi = \frac{1}{2V^2\,\overline{\psi}}\sum_{i=1}^{V}\sum_{j=1}^{V}\big|\psi^i - \psi^j\big|, \qquad \overline{\psi} = \frac{1}{V}\sum_{v=1}^{V}\psi^v. \tag{13}$$

A smaller I_ψ indicates more balanced contributions across views, whereas a larger I_ψ implies that only a few views dominate the collaborative clustering process.

Theorem 1 (Boundedness of the metric I_ψ) *Assume that for any $\mathcal{S}\subseteq\mathcal{V}\setminus\{v\}$, it holds that $\mathrm{UOT}(\mathbf{p},\mathbf{q}^{\mathcal{S}\cup\{v\}}) \le \mathrm{UOT}(\mathbf{p},\mathbf{q}^{\mathcal{S}})$. Equivalently, $E(\mathcal{S}\cup\{v\}) \ge E(\mathcal{S})$. If there exists an effective improvement such that $E(\mathcal{V}) > E(\emptyset)$ (i.e., $\sum_v \psi^v > 0$), then the contribution-imbalance index satisfies*

$$I_\psi \in \Big[0,\; 1-\tfrac{1}{V}\Big]. \tag{14}$$

The lower bound of 0 is achieved if and only if $\psi^1 = \cdots = \psi^V = \frac{1}{V}$. In contrast, the upper bound $1-\frac{1}{V}$ occurs in an extreme dominance scenario in which one view has $\psi^v = 1$ while all others are 0. (The proof is provided in Appendix A.)

Remark 1 *Since $\mathrm{UOT}(\cdot,\cdot) \ge 0$, the utility satisfies $E(\mathcal{C}) \in (0,1]$. If $E(\mathcal{V}) = E(\emptyset)$ (e.g., when all views provide no effective improvement to the fusion), then by (12) we have $\sum_v \phi^v = 0$, and consequently (13) becomes undefined. In practice, a degenerate convention $\psi^v = \frac{1}{V}$ can be adopted, which yields $I_\psi = 0$, consistent with the intuition of "no effective contribution disparity."*

3.5. Semantic Clustering with Fused Distributions

Based on Eq.(4), we obtain the fused distribution that integrates information from all views, and perform clustering directly within this shared representation space. For each sample i, we derive its soft cluster assignment q_{ij}, and assign the cluster label as $s_i = \arg\max_j q_{ij}$.

Following the strategy in [39], the clustering loss is defined as the Kullback–Leibler (KL) divergence between the target distribution $\mathbf{P}$ and the predicted distribution $\mathbf{Q}$, where the target distribution $\mathbf{P}$ is derived from $\mathbf{Q}$ and defined as:

$$p_{ij} = \frac{q_{ij}^2/\sum_i q_{ij}}{\sum_{j'}\Big(q_{ij'}^2/\sum_i q_{ij'}\Big)}. \tag{15}$$

Accordingly, the clustering loss is formulated as:

$$\mathcal{L}_c = \mathrm{KL}(\mathbf{P}\|\mathbf{Q}) = \sum_{i=1}^{N}\sum_{j=1}^{K} p_{ij}\log\frac{p_{ij}}{q_{ij}}. \tag{16}$$

3.6. Objective Function

Based on the above definitions, our overall optimization objective consists of three components: the intra-view reconstruction loss $\mathcal{L}_{rec}$, the cross-view semantic consistency loss $\mathcal{L}_{ccl}$, and the clustering loss $\mathcal{L}_c$. In general, the total objective function is formulated as follows:

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda_1\mathcal{L}_{ccl} + \lambda_2\mathcal{L}_c, \tag{17}$$

where λ_1 and λ_2 are balancing hyperparameters used to control the relative importance of cross-view consistency and clustering objectives, respectively.

3.7. Contribution-Balanced Enhancement Strategy

In imbalanced incomplete multi-view scenarios, different views exhibit significant disparities in data completeness and information quality. If optimized uniformly, the model tends to overfit a few dominant (high-contribution) views at early stages, while weak views (low-contribution) remain underutilized due to unstable representations and noisy gradients. To address this issue, we propose a contribution-balanced enhancement strategy instantiated via a View-Adaptive Curriculum Learning (VACL) mechanism, which dynamically adjusts view-wise optimization weights based on their contributions. This design enables a gradual transition from strong-view-dominated optimization to balanced multi-view collaboration.

Curriculum Weight Design. Let ψ^v denote the Shapley contribution of the v-th view, reflecting its relative importance at the current training stage. We define the curriculum factor as:

$$\gamma_v(t) = \frac{t}{T_c}\Big[(1-\psi^v)^2 + \varepsilon\Big], \tag{18}$$

where T_c represents the curriculum period (typically 30% of total epochs), and ε is a small constant to prevent numerical underflow. The time-dependent term $\frac{t}{T_c}$ increases linearly with training progress, implementing a progressive curriculum schedule, while the squared term $(1-\psi^v)^2$ emphasizes weaker views (i.e., those with smaller ψ^v) during the curriculum stage.

Optimization Objective and Parameter Update. Let $\theta_t^{(v)}$ denote the parameters of the v-th view at iteration t, and $\mathcal{L}^{(v)}(\theta_t^{(v)})$ the corresponding loss. The joint objective at iteration t is formulated as:

$$\mathcal{L}_t = \sum_{v=1}^{V}\gamma_v(t)\,\mathcal{L}^{(v)}(\theta_t^{(v)}). \tag{19}$$

Following the gradient descent rule, parameters are updated as:

$$\theta_{t+1}^{(v)} = \theta_t^{(v)} - \eta\,\gamma_v(t)\,\nabla_{\theta_t^{(v)}}\mathcal{L}^{(v)}(\theta_t^{(v)}), \tag{20}$$

where η denotes the global learning rate. The effective gradient for the v-th view can thus be written as:

$$g_t^{(v)} = \gamma_v(t)\,\nabla_{\theta_t^{(v)}}\mathcal{L}^{(v)}(\theta_t^{(v)}). \tag{21}$$

As training proceeds and weak views remain with lower ψ_v, their corresponding $\gamma_v(t)$ values increase, thereby amplifying their gradient responses and promoting balanced convergence across views.

Remark 2 *The proposed curriculum factor $\gamma_v(t)$ serves as a smooth, bounded, and monotonic modulation mechanism. This view-adaptive curriculum learning strategy enables a gradual transition from strong-view dominance to balanced multi-view collaboration through dynamic time-dependent scheduling, while ensuring the Lipschitz stability and numerical reliability of weight updates. (see Appendix B for the proof).*

4. Experiments

4.1. Datasets and Experimental Setting

Table 1. Information of five datasets.

Dataset	Samples	Views	Dimensionality
Synthetic3d	600	3	3/3/3
Reuters	1200	5	4819/4810/4892/4858/4777
Caltech101	1474	6	48/40/254/1984/512/928
Wikipedia	2866	2	128/10
Animal	10158	2	4096/4096

Datasets and Data pre-processing We conduct experiments on five benchmark multi-view datasets, namely Synthetic3d, Reuters, Caltech101, Wikipedia, and Animal. The detailed statistics of these datasets are summarized in Table 1. To realistically emulate imbalanced missing conditions in multi-view data, we generate incomplete datasets with controlled randomness and imbalance from the originally complete ones. An average missing rate $r \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ is specified to regulate the global incompleteness level. For each view v, a random missing factor α_v is sampled under the constraint $\sum_v \alpha_v = V$, ensuring that the overall average missing rate remains r. The view-specific missing rate is then defined as $r_v = \alpha_v \times r$, producing heterogeneous missing patterns across views. Moreover, each sample is guaranteed to be observable in at least one view to avoid completely missing instances.

Implementation Details We evaluate the proposed model using three standard clustering metrics: ACC, NMI, and ARI. All experiments are optimized with the Adam optimizer. The training process includes two stages with 100

Table 2. The clustering performance comparisons on five imbalanced incomplete multi-view datasets with varying missing rates. The best results are highlighted in **bold**, while the second best results are underlined. The missing rate of each view is defined as $r_v = \alpha_v \times r$.

Dataset	Method	Missing Rate $r = 0.1$			Missing Rate $r = 0.2$			Missing Rate $r = 0.3$			Missing Rate $r = 0.4$			Missing Rate $r = 0.5$		
		ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
Synthetic3d	CPSPAN	0.9350	0.8433	0.8883	0.8717	0.6974	**0.7967**	0.8217	0.5823	0.5822	0.8200	0.5302	0.5301	0.6067	0.3013	0.3181
	RPCIC	0.9167	0.7206	0.7694	0.8650	0.6560	0.6595	0.6900	0.4647	0.4268	0.7000	0.5443	0.4805	0.6467	0.3074	0.3016
	GHICMC	0.9600	0.8370	0.8843	0.9000	0.6889	0.6834	0.8525	0.5356	0.5892	0.8417	0.5259	0.5781	**0.6817**	0.4103	0.3804
	PMIMC	0.9533	0.8421	0.8716	0.9017	0.6882	0.6890	0.8417	0.5891	0.5818	0.6867	0.5110	0.4659	0.6533	0.3021	0.2952
	FREECSL	0.7717	0.4877	0.4933	0.6033	0.3429	0.3323	0.8150	0.4922	0.5366	0.6383	0.4346	0.4145	0.6017	0.3044	0.2957
	NBIMVC	0.9317	0.7464	0.8070	0.9100	0.7003	0.7594	0.8183	0.6098	0.5723	0.8567	0.6161	0.6066	0.6550	0.3088	0.3035
	BURG	0.8150	0.4922	0.5366	0.7717	0.4877	0.4933	0.7017	0.3544	0.3457	0.6383	0.4346	0.4145	0.6033	0.3029	0.3123
	ICER	**0.9650**	**0.8582**	**0.8988**	**0.9217**	**0.7392**	0.7822	**0.9133**	**0.7182**	**0.7616**	**0.9183**	**0.7341**	**0.7767**	0.6717	**0.4676**	**0.3998**
Reuters	CPSPAN	0.3517	0.1664	0.0990	0.3342	0.1546	0.0794	0.3233	0.1230	0.0613	0.3025	0.0858	0.0449	0.2617	0.0991	0.0361
	RPCIC	0.4208	0.2102	**0.2216**	0.4608	0.2602	0.2116	0.4008	**0.2202**	**0.1916**	0.3408	**0.1802**	**0.1216**	0.2808	0.1102	0.0716
	GHICMC	0.2367	0.1179	0.0125	0.2158	0.1005	0.0138	0.2458	0.0666	0.0335	0.2075	0.0806	0.0072	0.2008	0.0292	0.0022
	PMIMC	0.3458	0.1488	0.0786	0.3442	0.1311	0.0760	0.3158	0.0865	0.0618	0.2567	0.0539	0.0326	0.2308	0.0551	0.0302
	FREECSL	0.3858	0.2003	0.1583	0.4217	0.2074	0.1590	0.3775	0.1843	0.1088	0.2958	0.1051	0.0911	0.1855	0.1093	0.0515
	NBIMVC	0.3075	0.1148	0.0691	0.4017	0.1786	0.1335	0.3508	0.1417	0.0973	0.3450	0.1101	0.0862	0.3267	0.1238	0.0675
	BURG	0.3775	0.1825	0.1037	0.2858	0.1306	0.0730	0.1717	0.0112	0.0000	0.2892	0.1069	0.0698	0.2883	**0.1268**	0.0697
	ICER	**0.5300**	**0.2826**	0.2138	**0.5208**	**0.2729**	**0.2269**	**0.4325**	0.1849	0.1444	**0.3792**	0.1410	0.0976	**0.3625**	0.1177	**0.0846**
Caltech101	CPSPAN	0.3968	0.3469	0.3543	0.3430	0.2264	0.2234	0.3555	0.2623	0.1508	0.3064	0.1567	0.1777	0.2844	0.0924	0.1025
	RPCIC	0.6936	0.4545	0.4276	0.4491	0.1731	0.2774	0.3028	0.2532	0.1284	0.3548	0.1375	0.1761	0.3044	0.1021	0.1361
	GHICMC	0.5396	0.3766	0.3065	0.3946	0.2016	0.2694	0.3253	0.2594	0.1408	0.2740	0.2062	0.1336	0.2560	0.1003	0.1087
	PMIMC	0.3834	0.2910	0.1735	0.2892	0.1740	0.1900	0.3266	0.2747	0.1496	0.3736	0.2576	0.1841	0.2316	0.0965	0.1729
	FREECSL	0.3266	0.2747	0.1496	0.3535	0.1949	0.1709	0.3834	0.2910	0.1735	0.4144	0.2769	0.1826	0.4012	0.1127	0.1493
	NBIMVC	0.5153	0.6191	0.3956	0.4470	0.1377	0.2359	0.4491	0.2731	0.2774	0.4504	0.2317	0.2740	0.2188	0.0916	0.1013
	BURG	0.4788	0.4457	0.2155	0.4556	0.1563	0.2085	0.3946	0.2016	0.1694	0.4152	0.2264	0.2023	0.3852	0.1019	0.1532
	ICER	**0.7035**	**0.6401**	**0.6159**	**0.6221**	**0.4727**	**0.4393**	**0.5109**	**0.4377**	**0.3548**	**0.5312**	**0.4183**	**0.3626**	**0.4288**	**0.2826**	**0.2303**
Wikipedia	CPSPAN	0.1825	0.0483	0.0194	0.1790	0.0543	0.0226	0.1748	0.0463	0.0189	0.1821	0.0530	0.0216	0.1117	0.0330	0.1004
	RPCIC	0.1936	0.0576	0.0266	0.1923	0.0593	0.0284	0.1839	0.0608	0.0274	0.1891	0.0578	0.0277	0.1825	0.0517	0.0235
	GHICMC	0.4525	0.3418	0.2669	0.4368	0.3383	0.2619	0.4013	0.3018	0.2154	0.3747	0.2808	0.1949	0.3873	0.2883	0.2056
	PMIMC	0.2041	0.0667	0.0324	0.1783	0.0508	0.0225	0.1832	0.0564	0.0240	0.1706	0.0515	0.0193	0.1856	0.0553	0.0245
	FREECSL	0.4529	0.3859	0.2736	0.3304	0.2457	0.1450	0.3535	0.2949	0.1709	0.3095	0.2458	0.1538	0.4170	0.3134	0.2217
	NBIMVC	0.5286	0.4504	0.3430	0.4491	0.3731	0.2774	0.4470	0.3377	0.2359	0.2404	0.1054	0.0344	0.1525	0.0163	0.0107
	BURG	0.5527	0.4905	0.3367	0.4979	**0.4556**	**0.3425**	0.4934	0.3788	0.2696	0.4096	0.3222	0.1431	0.3829	0.3154	0.1035
	ICER	**0.5684**	**0.4945**	**0.3968**	**0.5098**	0.4228	0.3385	**0.4972**	**0.3796**	**0.3073**	**0.4669**	**0.3367**	**0.2607**	**0.4546**	**0.3713**	**0.2311**
Animal	CPSPAN	0.6117	0.6942	0.5096	0.5011	0.6249	0.3975	0.4325	0.5464	0.2158	0.4371	0.5647	0.2126	0.3231	0.4788	0.2214
	RPCIC	0.5084	0.5837	0.3884	0.4517	0.5172	0.3141	0.3430	0.4264	0.2234	0.2657	0.3668	0.1588	0.2557	0.3575	0.1406
	GHICMC	0.5279	0.6449	0.4225	0.5034	0.6214	0.3950	0.5153	0.6191	0.2556	0.4756	**0.6005**	0.2035	0.4502	0.5625	**0.3483**
	PMIMC	0.4363	0.5483	0.3174	0.3953	0.4971	0.2749	0.3946	0.5016	0.2394	0.3818	0.4992	0.2226	0.3448	0.4863	0.2299
	FREECSL	0.4282	0.5416	0.3126	0.4201	0.5256	0.3023	0.3825	0.5138	0.2494	0.4017	0.4925	0.2415	0.3230	0.4115	0.2065
	NBIMVC	0.3647	0.5125	0.2563	0.3492	0.4587	0.2202	0.2773	0.3946	0.1620	0.1151	0.1656	0.0412	0.0402	0.0083	0.0000
	BURG	0.5631	0.6679	0.4624	0.5290	0.6274	0.4272	0.4385	0.5800	0.2385	0.4507	0.5826	0.3288	0.3992	0.5383	0.3266
	ICER	**0.6240**	**0.7034**	**0.5294**	**0.5854**	**0.6713**	**0.4753**	**0.5471**	**0.6373**	**0.4088**	**0.5210**	0.5943	**0.3646**	**0.4628**	**0.5635**	0.3210

and 300 epochs, respectively, using a batch size of 256. All implementations are based on PyTorch 2.4.0 and run on a workstation with an NVIDIA GeForce RTX 3090 GPU and 64 GB RAM.

4.2. Compared Methods and Main Results

To rigorously assess our method, we benchmark it against seven state-of-the-art baselines: CPSPAN[13], RPCIC[46], GHICMC[1] PMIMC[47], FREECSL[3], NBIMVC[43], BURG[14].

Table 2 reports the clustering results of all compared methods on five imbalanced incomplete multi-view datasets under different missing rates. The main observations are as follows:

- ICER achieves the best overall performance. It consistently delivers the top results on Caltech101 and Animal across all missing rates and metrics. For instance, on Animal with $r = 0.1$, ICER reaches an ACC of 0.6240, outperforming the strongest baseline CPSPAN (0.6117). On Caltech101 with $r = 0.2$, ICER improves ACC from 0.4556 (BURG) to 0.6221 and NMI from 0.2264 (CPSPAN) to 0.4727, showing clear advantages over competing methods.
- ICER maintains superior performance across diverse datasets. Beyond Caltech101 and Animal, ICER also acquires good results on Reuters, Synthetic3d, and Wikipedia under missing rates. Several deep baselines degrade noticeably when missingness becomes more imbalanced, while ICER maintains stable performance across all datasets, indicating its ability to alleviate collaboration bias caused by uneven view missingness.
- ICER demonstrates strong robustness under high missing

rates. Although all methods degrade as the missing rate increases, ICER maintains clear advantages and remains the best-performing method at $r = 0.5$ across all datasets. For instance, on Wikipedia, its ACC decreases moderately from 0.5684 to 0.4546, while on Animal it still achieves the best ACC of 0.4628 at $r = 0.5$. On Reuters with $r = 0.5$, ICER attains the highest ACC (0.3625), outperforming the closest competitor NBIMVC.

4.3. Ablation study

We report ACC, NMI, and ARI as evaluation metrics.

Table 3. Ablation study on Synthetic3d, Caltech101 and Animal. We report three evaluation metrics result.(The missing rate is 0.3)

Datasets	$\mathcal{L}_{rec}$	$\mathcal{L}_{ccl}$	$\mathcal{L}_c$	ACC	NMI	ARI
Synthetic3d	✓			0.4283	0.0563	0.0296
	✓	✓		0.3350	0.0111	0.0007
	✓		✓	0.7433	0.5470	0.4789
	✓	✓	✓	**0.9133**	**0.7182**	**0.7616**
Caltech101	✓			0.3806	0.2179	0.1554
	✓	✓		0.4213	0.0471	0.0263
	✓		✓	0.4853	0.3713	0.3019
	✓	✓	✓	**0.5109**	**0.4377**	**0.3548**
Animal	✓			0.4725	0.5166	0.3230
	✓	✓		0.4740	0.5191	0.4152
	✓		✓	0.5434	0.6367	0.4073
	✓	✓	✓	**0.5471**	**0.6373**	**0.4088**

Table 3 evaluates the contributions of the three loss components. Using only $\mathcal{L}_{rec}$ yields the weakest results because it cannot resolve cross-view inconsistency under imbalanced missingness. Adding $\mathcal{L}_{ccl}$ noticeably improves performance by enhancing semantic alignment, while the full model with $\mathcal{L}_c$ further strengthens discriminative clustering and achieves the best results, confirming that the three losses are complementary.

Table 4. Ablation study on whether to perform VACL on three datasets with the missing rate fixed at 0.5.

Method	Wikipedia		Caltech101		Animal	
	ACC	I_ψ	ACC	I_ψ	ACC	I_ψ
VACL	0.4546	0.0020	0.4288	0.0148	0.4628	0.0490
VACL_without	0.3699	0.2700	0.4072	0.0225	0.4339	0.0681

Table 4 and Fig. 3 further analyze the impact of VACL. When VACL is removed, the imbalance metric I_ψ increases substantially across all datasets, indicating that certain views dominate the learning process. After introducing VACL, I_ψ is significantly reduced and ACC consistently improves on Wikipedia, Caltech101, and Animal. Moreover, Fig. 3 shows that VACL leads to a more uniform distribution of view contributions. These results demonstrate that VACL effectively mitigates collaboration bias and enhances clustering performance.

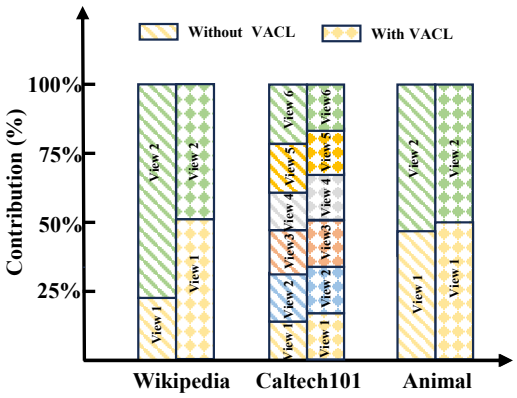
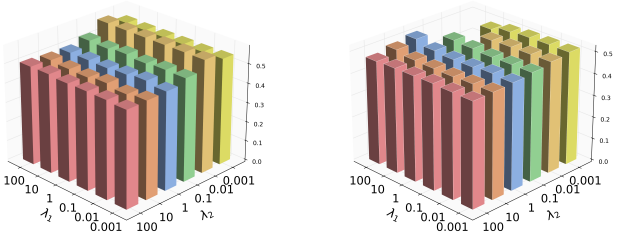


Figure 3. Comparison of view contribution distributions with and without VACL under a missing rate of 0.5.

4.4. Sensitivity Analysis

We further conducted a parameter sensitivity analysis on the Animals and Wikipedia datasets under a fixed missing rate of 0.3, as shown in Figure 4. The model involves two trade-off hyperparameters, λ_1 and λ_2, which were varied within the range $\{0.001, 0.01, 0.1, 1, 10, 100\}$. For both datasets, the clustering performance remains relatively stable as the two parameters change, showing only minor fluctuations across different combinations of λ_1 and λ_2. This consistency indicates that our method is insensitive to hyperparameter tuning and maintains robust clustering performance under a wide range of settings.



(a) ACC on Animal (b) ACC on Wikipedia

Figure 4. Sensitivity analysis of $\lambda 1$ and $\lambda 2$ for our method over Animal and Wikipedia (The missing rate is 0.3).

5. Conclusion

This paper addresses view-contribution imbalance in incomplete multi-view clustering by combining Shapley-value-based contribution attribution with unbalanced optimal transport to evaluate cross-view discrepancies, constructing a view contribution imbalance metric, and introducing a view-adaptive curriculum learning strategy to strengthen weak views. Extensive experiments on multiple benchmarks demonstrate consistent improvements over state-of-the-art methods.

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